

KPI & Metrics Framework

University Course Feedback Analytics Platform

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Executive Summary

This document defines 8 critical Key Performance Indicators (KPIs) for measuring the success of the feedback analytics platform. Each KPI includes:

- Clear definition and measurement approach
- Target value (aspirational goal)
- Alert threshold (action required)
- Frequency of measurement
- Who owns the metric
- How the metric is calculated

These KPIs measure three dimensions:

- Teaching Quality - Student satisfaction with instruction
- System Effectiveness - Quality and reliability of the platform
- Organizational Impact - Whether the system drives actual improvements

Core Teaching Quality KPIs

KPI 1: Course Satisfaction Rating

Definition: Average rating of the question "Overall, how satisfied are you with this course?" on a 1-5 scale.

Calculation:

Course Satisfaction = $\text{SUM}(\text{all ratings for question}) / \text{COUNT}(\text{ratings})$
Range: 1.0 - 5.0

Example:

Course: Database Design
Ratings: 5, 4, 5, 3, 4, 5, 4, 4
Calculation: $(5+4+5+3+4+5+4+4) / 8 = 34 / 8 = 4.25$

Target Value: ≥ 4.0 (80% of students satisfied or very satisfied)

Alert Threshold: < 3.5 (below 70% satisfaction; requires intervention)

Measurement Frequency: Real-time (calculated as new feedback arrives); aggregated monthly

Ownership: Faculty member (for their course); Department Chair (for department aggregate)

How to Read:

4.5-5.0: Excellent teaching; celebrate and share practices
4.0-4.5: Good teaching; room for incremental improvement
3.5-4.0: Satisfactory but concerning; intervention recommended
3.0-3.5: Below acceptable threshold; urgent intervention needed
<3.0: Serious issues; escalate to department leadership

Context & Notes:

This is the PRIMARY KPI for teaching quality
Compare semester-to-semester to track improvement/decline
Compare across courses and instructors (with context—different course levels have different satisfaction)
Use dashboard heatmap to visualize trends over time

KPI 2: Content Clarity Rating

Definition: Average rating of the question "The instructor clearly explained course content" on a 1-5 scale.

Calculation:

Content Clarity = $\text{SUM}(\text{ratings on clarity question}) / \text{COUNT}(\text{ratings})$
Range: 1.0 - 5.0

Target Value: ≥ 4.0 (students generally agree content was clear)

Alert Threshold: < 3.5 (students report confusion on course content)

Measurement Frequency: Real-time; aggregated monthly and semester-end

Ownership: Faculty member (for self-reflection); Department Chair (for identifying PD needs)

Interpretation Tips:

If Content Clarity $<$ Overall Satisfaction: students like instructor but find content difficult; may need better scaffolding

If Content Clarity $>$ Overall Satisfaction: content is clear but something else is off (pace, engagement, grading fairness); investigate comments

Compare to own historical data to track if changes improved clarity

Action Examples:

Score drops 0.3 points semester-over-semester: "I changed textbooks; that might have affected clarity. Let's gather more feedback."

Consistently 3.8: "I'm doing pretty well but there's room. I'll add more worked examples."

Drops below 3.5: "Multiple students say unit 4 is confusing. I'll redesign that unit for next semester."

KPI 3: Instructor Engagement Rating

Definition: Average rating of the question "The instructor was engaging and responsive" on a 1-5 scale.

Calculation:

Instructor Engagement = $\text{SUM}(\text{ratings on engagement question}) / \text{COUNT}(\text{ratings})$
Range: 1.0 - 5.0

Target Value: ≥ 4.0 (students perceive instructor as engaged and responsive)

Alert Threshold: < 3.5 (students feel instructor is disengaged or unavailable)

Measurement Frequency: Real-time; aggregated monthly and semester-end

Ownership: Faculty member (for self-improvement); Department Chair (for evaluating faculty)

How to Use:

High engagement rating (4.5+) despite moderate satisfaction: students like the instructor but course may have content/structure issues

Low engagement (< 3.5): escalate to faculty as urgent issue

Track over years: is engagement declining with age/experience? may signal burnout

KPI 4: Sentiment Score

Definition: Percentage of feedback with positive sentiment (determined by automated sentiment analysis) out of all feedback.

Calculation:

Sentiment Score = $(\text{COUNT}(\text{feedback with positive sentiment}) / \text{TOTAL COUNT}(\text{feedback})) \times 100$
Range: 0% - 100%

Example:

Course: Web Technologies

Total feedback: 50 comments

Positive sentiment: 38 comments

Negative sentiment: 10 comments

Neutral sentiment: 2 comments

Sentiment Score = $(38 / 50) \times 100 = 76\%$

Target Value: $\geq 75\%$ (majority of feedback positive or constructive)

Alert Threshold: $< 60\%$ (more negative than positive feedback; concerning)

Measurement Frequency: Real-time (updated as new feedback arrives); aggregated daily

Ownership: Department Chair (system-wide); Faculty (individual course)

Sentiment Categories:

Positive: Comments praising instruction, course content, improvements, engagement
Examples: "Excellent pacing," "Clear explanations," "Responsive to student concerns"

Negative: Comments criticizing teaching, clarity, engagement, responsiveness
Examples: "Too fast-paced," "Confusing lectures," "Never responded to emails"

Neutral: Factual observations or mixed feedback
Examples: "Lots of homework," "Depends on the topic"

Sentiment Accuracy:

System uses automated sentiment analysis (TextBlob/VADER/transformer model)
Accuracy typically 85-90% on educational feedback
Flag outliers (e.g., very negative single comment) for manual review
Dashboard shows confidence score (0-100%) for each comment's sentiment

How to Interpret:

90%+ positive: Excellent; share teaching practices
75-90% positive: Good; some concerns to address
60-75% positive: Below target; analyze negative comments to identify issues
<60% positive: Alert; urgent intervention; typically paired with low satisfaction rating

KPI 5: Response Rate

Definition: Percentage of enrolled students who submitted feedback.

Calculation:

Response Rate = $(\text{COUNT}(\text{feedback submissions}) / \text{COUNT}(\text{enrolled students})) \times 100$
Range: 0% - 100%

Example:

Course: Database Design
Enrolled students: 120
Feedback submissions: 78
Response Rate = $(78 / 120) \times 100 = 65\%$

Target Value: $\geq 60\%$ (solid representation of student opinions; statistically meaningful)

Alert Threshold: $< 40\%$ (sample may not be representative; feedback may be biased)

Measurement Frequency: Ongoing during feedback window; aggregated at semester end

Ownership: Department Chair (encouraging participation); Course Instructor (through student messaging)

Context:

Higher response rates = more representative feedback = higher confidence in data

Lower response rates = may be skewed toward extreme opinions (very satisfied or very unsatisfied)

Response rate varies by:

Course level (intro courses often lower; upper-level courses higher)

Timing (earlier deadline → higher response; last-minute deadline → lower response)

Incentives (optional survey → ~40%; incentivized survey → ~65%)

Student motivation (final exams period → lower; earlier in semester → higher)

Improving Response Rate:

Send reminder emails (typically +10-15% increase)

Offer incentive (extra credit point, entry in raffle; typically +15-25% increase)

Make survey mobile-first (easier to complete on phone; typically +5-10%)

Reduce survey length (fewer questions = higher completion; ~5-minute target)

Emphasize impact ("Your feedback directly improves this course"; typically +10%)

How to Interpret:

70%+ response rate: Excellent engagement; can confidently act on feedback

60-70% response rate: Good; sufficient for decision-making

40-60% response rate: Acceptable but caution; may be biased toward extremes

<40% response rate: Concerning; consider oversampling before making major decisions

KPI 6: Feedback Velocity

Definition: Average number of days between receipt of feedback and faculty response/acknowledgment.

Calculation:

Feedback Velocity = $\text{SUM}(\text{days to response for all feedback}) / \text{COUNT}(\text{feedback})$

Range: 1 - ∞ days (typically 1-30 days)

Example:

Feedback received: Monday

Faculty reviews feedback: Wednesday (2 days)

Faculty responds/acknowledges in class: Thursday (3 days total)

Average across 50 comments = 4.2 days

Target Value: ≤ 7 days (faculty responds quickly; shows they're paying attention)

Alert Threshold: > 14 days (too slow; feedback momentum lost; students perceive inaction)

Measurement Frequency: Real-time; aggregated monthly

Ownership: Faculty member (responsibility to respond); Department Chair (monitoring)

How "Response" is Measured:

Direct Response: Faculty replies to comment in system (best)

Acknowledgment: Faculty mentions feedback in class ("Several students said X; here's how I'm addressing it")

Action: Faculty implements change (most impactful)

Documentation: Faculty marks comment as "reviewed" in system

Context:

Not all feedback needs individual response (some is general opinion)

Key feedback should be responded to (critical issues, specific suggestions)

Public/class-level responses more impactful than individual replies

Timeliness signals to students: "I'm listening and taking this seriously"

Why This Matters:

Closes feedback loop: students see their input matters

Maintains momentum: responding quickly shows engagement

Demonstrates accountability: faculty are responsive

Improves satisfaction: students appreciate acknowledgment

KPI 7: System Uptime

Definition: Percentage of time the feedback platform is available and functioning (available ÷ total time).

Calculation:

System Uptime = $((\text{Total Time} - \text{Downtime}) / \text{Total Time}) \times 100$

Measured over: Monthly, Quarterly, Annually

Example:

Month: 30 days = 43,200 minutes

Unplanned downtime: 60 minutes

System Uptime = $((43,200 - 60) / 43,200) \times 100 = 99.86\%$

Target Value: 99.5% (allows ~4 hours downtime per month; ~2 days per year)

Alert Threshold: < 99% (more than 7 hours downtime per month)

Measurement Frequency: Continuous automated monitoring; reported daily and monthly

Ownership: IT Department (system administration); System Owner (oversight)

Why This Matters:

Feedback collection window is limited (typically 2-3 weeks at semester end)

Downtime during this window blocks student feedback submission

Reliability = trust; if system is down, faculty/students lose confidence

Impacts ability to make semester-end decisions on course improvements

SLA (Service Level Agreement):

Target uptime during peak periods (last week of semester): 99.9%

Standard uptime target: 99.5%

Acceptable downtime:

Planned maintenance: 4 hours/quarter (scheduled off-hours)

Unplanned: <5 minutes/month

Emergency fixes: 30 minutes/quarter

KPI 8: Data Quality Rate

Definition: Percentage of feedback records that have complete and valid data (valid ratings and no critical errors).

Calculation:

Data Quality Rate = $(\text{COUNT}(\text{valid feedback records}) / \text{TOTAL COUNT}(\text{feedback records})) \times 100$

Where "valid" means:

- Rating is numeric and in range [1, 5]
- Feedback not flagged as spam/inappropriate
- No critical data corruption
- Course ID and timestamp present

Example:

Total feedback records: 1,000

Records with invalid ratings: 3

Records flagged as spam: 2

Records with missing timestamp: 1

Valid records: $1,000 - 3 - 2 - 1 = 994$

Data Quality Rate = $(994 / 1,000) \times 100 = 99.4\%$

Target Value: 100% (or $\geq 99.5\%$ allowing for 1-2 manual corrections)

Alert Threshold: < 99% (indicates potential system issues)

Measurement Frequency: Real-time validation; aggregated daily

Ownership: Data Team (validation); System Owner (monitoring)

Data Quality Checks:

Completeness: Required fields present (rating, course, timestamp)

Validity: Rating values in [1, 5]; timestamp is valid date

Uniqueness: Duplicate submissions detected and flagged

Consistency: Course ID exists in course master data; Faculty ID valid

Accuracy: Sentiment analysis confidence >60% (or manually review low-confidence)

How to Maintain High Quality:

Frontend validation: reject submissions with invalid ratings before sending

Backend validation: catch any malformed data before storing

Regular audits: monthly data quality checks

Monitoring: alerts on quality drops below 99%

Correction workflow: process for fixing erroneous records

Supplemental Metrics (Trending & Comparison)

KPI 9: Semester-Over-Semester Improvement

Definition: Change in average satisfaction rating from one semester to the next (for same course/instructor).

Calculation:

SoS Improvement = Current Semester Avg - Previous Semester Avg

Can be expressed as: Absolute change (points) or Percentage change (%)

Example:

- Spring 2025 satisfaction: 3.8

- Fall 2025 satisfaction: 4.1

- SoS Improvement = $4.1 - 3.8 = +0.3$ points (7.9% improvement)

Target Value: ≥ 0 (maintain or improve from previous semester)

Alert Threshold: < -0.2 (significant decline; investigate why)

Measurement Frequency: Semester-end (calculated after all feedback collected)

Ownership: Faculty (self-improvement tracking); Department Chair (performance monitoring)

KPI 10: Department Comparative Performance

Definition: How a department's average teaching quality compares to the institutional average and peer departments.

Calculation:

Department Average = AVG(all courses in department)

Institutional Average = AVG(all courses university-wide)

Comparative Index = $(\text{Department Average} / \text{Institutional Average}) \times 100$

If index = 100: At institutional average

If index = 105: 5% above institutional average (strong)

If index = 95: 5% below institutional average (needs improvement)

Example:

Department (Computer Science): 4.15

Institution Average: 4.08

Comparative Index = $(4.15 / 4.08) \times 100 = 101.7\%$

→ Computer Science is 1.7% above institutional average

Target Value: $\geq 100\%$ (meet or exceed institutional average)

Alert Threshold: $< 95\%$ (more than 5% below average; department needs support)

Measurement Frequency: Semester-end and annual

Ownership: Dean (comparative performance); Department Chair (department improvement)

KPI Monitoring & Governance

KPI Review Schedule

Frequency	Metrics Reviewed	Owner	Action
Real-Time	System Uptime, Data Quality	IT Department	Alert on threshold breach
Daily	Response Rate progress, Alerts triggered	Faculty	Acknowledge alerts
Weekly	Satisfaction trends (week-over-week)	Faculty, Chairs	Monitor for issues
Monthly	Satisfaction, Sentiment, Velocity (aggregated)	Department Chairs	Identify trends, plan interventions
Semester-End	All KPIs, SoS Improvement, Comparative Performance	Dean, Chairs	Strategic planning, resource allocation
Annual	Multi-year trends, benchmark comparisons	Dean, Provost	Strategic planning, accreditation

Alert Thresholds & Escalation

KPI	Alert Threshold	Action	Escalate If
Course Satisfaction	< 3.5	Faculty + Chair notified; review comments	< 3.0 or declining for 2+ semesters
Sentiment Score	< 60%	Analyze negative comments	< 50% or alert on 2+ courses
Response Rate	< 40%	Encourage more submissions; note potential bias	< 25% or trend declining
Feedback Velocity	> 14 days	Reminder to faculty to respond	> 21 days or pattern across courses
System Uptime	< 99%	IT investigates; log incident	< 98% or during peak period
Data Quality	< 99%	Data team reviews; identifies root cause	< 95% or critical data corruption

KPI Dashboard Views

Faculty View:

My Course Satisfaction (with trend chart)
My Sentiment Score (with positive/negative breakdown)
Recent Feedback Comments (latest 10, with sentiment tags)
My Response Rate to Feedback
Semester-over-Semester Comparison (my course)
Peer Comparison (my department average)

Department Chair View:

Department Satisfaction (all courses, heatmap by semester)
Department Sentiment Score (aggregate)
Response Rate across all courses
Alert Panel (courses below thresholds)
Comparative Performance (vs. institution)
SoS Improvement (which courses improved, which declined)

Dean View:

Institution Satisfaction (all departments)
Institution Sentiment Score
Department Comparative Performance
Multi-Year Trends (3-5 years)
Alert Panel (departments, system health)
KPI Progress vs. Targets

Formula Reference & Calculations

Basic KPI Formulas for Spreadsheet Implementation

=== Course Satisfaction ===

=AVERAGE(D2:D100)

=== Sentiment Score ===

=COUNTIF(E2:E100,"positive") / COUNTA(E2:E100) * 100

=== Response Rate ===

=COUNTA(A2:A100) / 120 * 100

=== Feedback Velocity ===

=AVERAGE(F2:F100)

=== SoS Improvement ===

=AVERAGE(C2:C100) - AVERAGE(B2:B100)

=== Department Comparative Index ===

=(AVERAGE(all_dept_courses) / AVERAGE(all_university_courses)) * 100

=== Data Quality Rate ===

=COUNTIF(status,"valid") / COUNTA(status) * 100

Target Setting Rationale

Why 4.0 for Satisfaction?

4.0 = 80% satisfaction; industry benchmark

Below 4.0 = only moderate satisfaction

Why 75% for Sentiment?

3:1 positive to negative ratio

Why 60% Response Rate?

Ensures statistical reliability

Why 7 Days for Feedback Velocity?

Maintains engagement and relevance

Success Indicators

KPI	Target	Success Indicator
Course Satisfaction	≥ 4.0	Students satisfied
Content Clarity	≥ 4.0	Students understand content
Instructor Engagement	≥ 4.0	Students feel engaged
Sentiment Score	$\geq 75\%$	Feedback mostly positive
Response Rate	$\geq 60\%$	Good participation
Feedback Velocity	≤ 7 days	Quick responses
System Uptime	$\geq 99.5\%$	Reliable system
Data Quality Rate	$\geq 99.5\%$	Accurate data

The End